

Introduction to Machine Learning

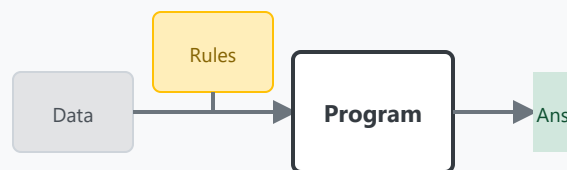
1. The Intuition: What is "Learning"?

At its core, Machine Learning is about using **examples** to infer rules about the world, rather than hard-coding them.

Imagine you want to teach a computer to recognize a cat.

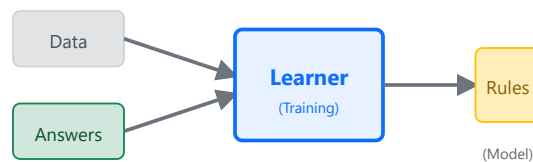
- **Traditional Programming (Top-Down):** You write rules: "If it has triangular ears, whiskers, and meows, it's a cat." This is hard because there are infinite variations.
- **Machine Learning (Bottom-Up):** You show the computer 1,000 pictures of cats and 1,000 pictures of dogs. The computer figures out the patterns itself.

Traditional Programming



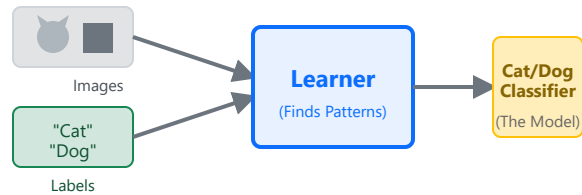
Rules + Data → Answers

Machine Learning



1. General Concept: Data + Answers → Rules

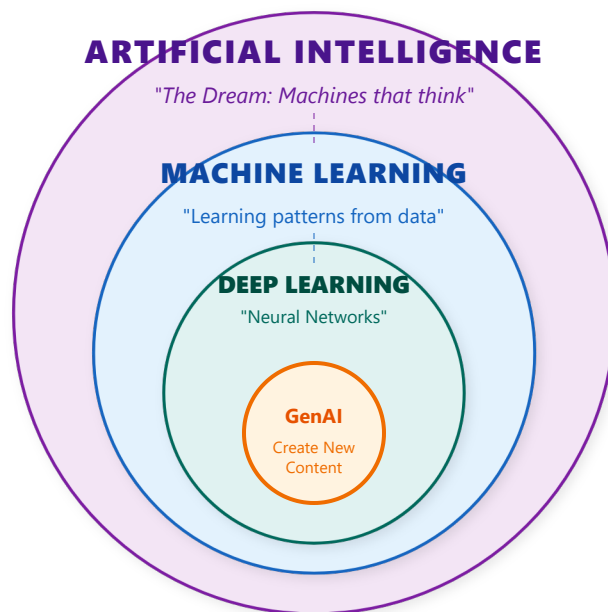
Example: Cat vs Dog



2. Specific Example: Images + Labels → Cat/Dog Model

The AI Landscape

How do the buzzwords fit together? A visual hierarchy.



AI

The broad science of mimicking human abilities.

ML

Algorithms that improve through experience.

DL

Multi-layered networks solving complex problems.

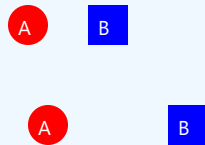
GenAI

Models that generate text, images, and code.

2. Supervised vs. Unsupervised Learning

There are two main ways we teach machines:

Supervised Learning



Data has LABELS (A vs B)

The teacher gives the answer key. "This image is a Cat, this is a Dog."

Unsupervised Learning



No LABELS (Just patterns)

No answer key. The computer finds structure on its own (e.g., "These items look similar").

Mathematical Formulation

Supervised Learning

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$$

- $x_i \in \mathbb{R}^d$: Input features (d-dimensional vector).
- $y_i \in \mathcal{Y}$: Target label (Output).

The Goal (Equation):

$$\hat{y} = f(x)$$

Learn a function to map Input x to Output y .

Unsupervised Learning

$$\mathcal{D} = \{x_i\}_{i=1}^n$$

- $x_i \in \mathbb{R}^d$: Input features only.
- No labels y_i provided.

The Goal (Equation):

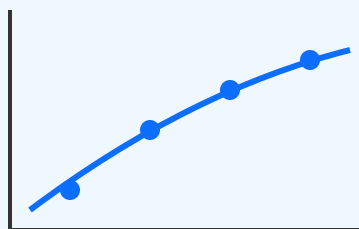
$$z = f(x)$$

Learn a function to map Input x to Hidden Structure z (e.g., Clusters).

3. Two Core Supervised Tasks: Regression vs Classification

Once we decide on supervised learning, we pick the output type:

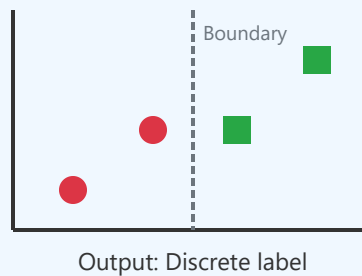
Regression



Output: Continuous number

Predict a real value: house price, temperature, stock value.

Classification



Predict a category: spam/not-spam, species A/B, fraud/legit.

Quick Math View

Regression

$$y \in \mathbb{R}$$

- Squared-error loss: $(y - \hat{y})^2$
- Goal: minimize average squared difference.

Classification

$$y \in \{0, 1, \dots, K-1\}$$

- 0-1 loss: $\mathbb{1}\{y \neq \hat{y}\}$
- Goal: minimize mis-classification rate.

4. Other Common Machine-Learning Tasks

Besides regression & classification, the ML toolbox includes:

Clustering – group similar items without labels (k-means, DBSCAN).

Example: Google Photos face grouping, Customer segmentation.

Dimensionality Reduction – compress high-D data while keeping structure (PCA, t-SNE).

Example: Image compression, Visualizing complex data.

Anomaly / Outlier Detection – flag rare events that deviate from the norm.

Example: Credit card fraud alerts, Cyber intrusion detection.

Ranking & Recommendation – order items or suggest products (collaborative filtering, learning-to-rank).

Example: Netflix/Spotify recommendations, TikTok/Instagram feeds.

Generative AI – create new content like text, images, or code (ChatGPT, Midjourney, GitHub Copilot).

Reinforcement Learning – an agent learns by trial-and-error rewards (Video Games, robotics).

All tasks still follow the same theme: extract patterns from data, generalize to the unseen.

3. The Math: Defining "Error" (Risk)

How do we know if our model is good? We measure the **Risk** (Error).

Population Risk (The True Error)

$$R(h) = \mathbb{P}(h(X) \neq Y)$$

In simple terms: The probability that our model h makes a wrong prediction on any random data point from the real world. We can never calculate this perfectly because we don't see "all" data.

Analogy: *If we deploy our Cat/Dog classifier to the internet, the Population Risk is the percentage of ALL cat/dog images in existence that it misclassifies (e.g., calling a Cat a "Dog").*

Empirical Risk (The Training Error)

$$\hat{R}_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{h(x_i) \neq y_i\}$$

In simple terms: The average number of mistakes the model makes on the **training data** we actually have.

- **n**: Number of training examples.
- **h(x_i)**: The model's prediction.
- **y_i**: The correct answer.
- **1{...}**: Indicator function. It is 1 if the condition inside is true (mistake), and 0 otherwise. Think of it as if/else in mathematics

Numerical Example: Suppose we have a dataset of 100 images ($n = 100$): 50 Cats and 50 Dogs.

- The model misclassifies 5 Cats (thinks they are Dogs).
- The model misclassifies 10 Dogs (thinks they are Cats).
- Total Mistakes = 5 + 10 = 15.

The Empirical Risk is $\hat{R}_n(h) = \frac{15}{100} = 0.15$ (or 15% error).

The Goal (ERM): We try to find the model $\hat{h}$ that minimizes the Empirical Risk, hoping it will also minimize the true Population Risk.